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Chalk line marking devices and methods

Abstract

Chalk line marking devices suitable for placing chalk lines at selected chalk line spacings on a surface may include a device housing having a housing interior. The housing interior may be sized and configured to contain a supply of a powdered chalk material. A chalk marking assembly slot may be in the device housing. A chalk marking assembly may be selectively extendable from the housing interior through the chalk marking assembly slot. The chalk marking assembly may include an elongated, flexible main marking member. An elongated, flexible guide member may be disposed in parallel, spaced-apart relationship to the main marking member. Multiple center marking members may span the main marking member and the guide member. The center marking members may be disposed at mark spacings with respect to each other. The mark spacings may correspond in length to the respective chalk line spacings.

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Background/Summary

FIELD

(1) Illustrative embodiments of the disclosure generally relate to devices and methods suitable for marking chalk lines on a surface in construction and other applications. More particularly, illustrative embodiments of the disclosure relate to chalk line marking devices which are suitable for placing chalk lines at selected chalk line spacings on a bottom plate to mark the positions of wall studs on the bottom plate in construction of a wall.

SUMMARY

(2) Illustrative embodiments of the disclosure are generally directed to chalk line marking devices

suitable for placing chalk lines at selected chalk line spacings on a surface. An illustrative embodiment of the chalk line marking devices may include a device housing. A housing interior may be in the device housing. The housing interior may be sized and configured to contain a supply of a powdered chalk material. A chalk marking assembly slot may be in the device housing. A chalk marking assembly may be selectively extendable from the housing interior through the chalk marking assembly slot. The chalk marking assembly may include an elongated, flexible main marking member. An elongated, flexible guide member may be disposed in parallel, spaced-apart relationship to the main marking member. Multiple center marking members may span the main marking member and the guide member. The center marking members may be disposed at mark spacings with respect to each other. The mark spacings may correspond in length to the respective chalk line spacings.

(3) In another aspect, a reel assembly may be disposed in the housing interior of the device housing and a reel crank may engage the reel assembly for rotation. The chalk marking assembly may be wound on the reel assembly.

(4) In another aspect, each of the main marking member and the guide member of the chalk marking assembly may be a segment of string, thread, twine, cord or a combination thereof.

(5) In another aspect, the mark spacings may include inch mark spacings.

(6) In another aspect, the mark spacings may include 24 inch mark spacings.

(7) In another aspect, at least one chalk opening may be in the device housing. At least one chalk opening door may be selectively deployable in open and closed door positions with respect to the chalk opening.

(8) In another aspect, at least one slot seal may be in the chalk marking assembly slot. The slot seal may sealingly engage the chalk marking assembly.

(9) In another aspect, at least one engaging member may be on the main marking member and the guide member of the chalk marking assembly.

(10) In another aspect, the center marking members may include multiple center marking member pairs.

(11) In some embodiments, the chalk line marking devices may include a chalk wheel handle. A chalk wheel may be supported by and journaled for rotation with respect to the chalk wheel handle. The chalk wheel may include a wheel housing including a circumferential outer wheel housing portion. Multiple chalk dispensing cavities may be in the outer wheel housing portion. Each chalk dispensing cavity may be sized and configured to contain a supply of a powdered chalk material and dispense the powdered chalk material from the wheel housing at the outer wheel housing portion responsive to rolling of the chalk wheel on the surface. The chalk dispensing cavities may be disposed at mark spacings with respect to each other along the outer wheel housing portion. The mark spacings may correspond in length to the respective chalk line spacings.

(12) In another aspect, multiple chalk dispensing inserts may be deployed in the respective chalk dispensing cavities. The chalk dispensing inserts may be sized and configured to contain the powdered chalk material and dispense the powdered chalk material from the wheel housing at the outer wheel housing portion.

(13) In another aspect, the chalk dispensing inserts may be releasably disposed in the respective chalk dispensing cavities.

(14) In another aspect, a wheel housing interior may be in the wheel housing. The wheel housing interior may be sized and configured to contain the powdered chalk material and dispense the powdered chalk material to the respective chalk dispensing cavities.

(15) In another aspect, at least one chalk opening may be in the wheel housing and communicate with the wheel housing interior. At least one chalk opening door may be selectively deployable in open and closed door positions with respect to the chalk opening.

(16) In another aspect, the mark spacings may include 16 inch mark spacings.

(17) In another aspect, the mark spacings may include 24 inch mark spacings.

(18) In another aspect, the chalk dispensing cavities may include multiple respective chalk dispensing cavity pairs.

(19) In some embodiments, the chalk line marking devices may include a chalk wheel handle. A chalk wheel may be supported by and journaled for rotation with respect to the chalk wheel handle. The chalk wheel may include a wheel housing including a pair of spaced-apart side wheel housing portions, a circumferential outer wheel housing portion on the side wheel housing portions and a wheel housing interior formed by the side wheel housing portions and the outer wheel housing portion. Multiple chalk dispensing cavities may be in the outer wheel housing portion. The chalk dispensing cavities may be disposed at mark spacings with respect to each other. The mark spacings may correspond in length to the respective chalk line spacings. Multiple inner cavity slots may establish communication between the wheel housing interior and the respective chalk dispensing cavities. Multiple outer cavity slots may communicate with the respective chalk dispensing cavities and open to the outer wheel housing portion. Multiple chalk dispensing inserts may be in the respective chalk dispensing cavities and communicate with the respective inner cavity slots and outer cavity slots. The chalk dispensing inserts may be sized and configured to contain a supply of powdered chalk and dispense the powdered chalk from the wheel housing through the respective outer cavity slots at the outer wheel housing portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Illustrative embodiments of the disclosure will now be described, by way of example, with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a front perspective view of an illustrative embodiment of the chalk line marking devices with a chalk marking assembly deployed in a partially extended position from a device housing of the device;
- (3) FIG. 2 is a top view of the illustrative chalk line marking device of FIG. 1;
- (4) FIG. 3 is a left side view, partially in section, of the illustrative chalk line marking device with the chalk marking assembly in the extended position;
- (5) FIG. 4 is a left side view of the illustrative chalk line marking device with the chalk marking assembly in the retracted position in the device housing;
- (6) FIG. 5 is a longitudinal sectional view, taken along section lines 5-5 in FIG. 2, of a typical device housing of the illustrative chalk line marking device;
- (7) FIG. 6 is an enlarged sectional view, taken at section line 6 in FIG. 5, illustrating a typical slot seal disposed in a chalk marking assembly slot in the device housing and the chalk marking assembly sealingly engaged by the slot seal as the chalk marking assembly extends through the chalk marking assembly slot from the device housing;
- (8) FIG. 7 is an enlarged sectional view of a top housing portion of the device housing with a chalk opening in the top housing portion and a typical chalk opening door deployed in a closed door position to close the chalk opening;
- (9) FIG. 8 is an enlarged sectional view of the top housing portion of the device housing with the chalk opening door deployed in an open door position for placement of powdered chalk (illustrated in phantom) into the device housing through the chalk opening;
- (10) FIG. 9 is a side view, partially in section, of a wall section of a building having a foundation and a bottom plate on the foundation, more particularly illustrating typical placement of chalk lines on the bottom plate at selected chalk line spacings with respect to each other in typical application of the chalk line marking devices;
- (11) FIG. 10 is a top view, partially in section, of the foundation and the wall section on the foundation of the building illustrated in FIG. 9, with the chalk lines formed on the bottom plate at

the selected chalk line spacings;

(12) FIG. 11 is a side view, partially in section, of the wall section of the building, with multiple wall studs extending upwardly from the bottom plate at the respective chalk lines and at wall stud spacings which correspond to the chalk line spacings of the chalk lines illustrated in FIG. 10;

(13) FIG. 12 is a side view of an illustrative chalk wheel embodiment of the chalk line marking devices;

(14) FIG. 13 is a front view of the illustrative chalk line marking device illustrated in FIG. 12;

(15) FIG. 14 is a sectional view, taken along section lines 14-14 in FIG. 13, of the chalk wheel of the illustrative chalk line marking device;

(16) FIG. 15 is a sectional view, taken along section lines 15-15 in FIG. 13, of the chalk wheel;

(17) FIG. 16 is an enlarged sectional view of the outer wheel housing portion of the wheel housing of the chalk wheel, more particularly illustrating a pair of spaced-apart chalk dispensing cavities in the outer wheel housing portion and a chalk dispensing insert in each chalk dispensing cavity;

(18) FIG. 17 is a top view of the chalk dispensing cavities with the chalk dispensing inserts (illustrated in phantom) deployed in place in the respective chalk dispensing cavities;

(19) FIG. 18 is an exploded top view of the chalk dispensing cavities with the chalk dispensing inserts removed from the respective cavities;

(20) FIG. 19 is a side view, partially in section, of a wall section of a building having a foundation and a bottom plate on the foundation, more particularly illustrating typical placement of chalk lines on the bottom plate at selected chalk line spacings with respect to each other in typical application of the chalk wheel embodiment of the chalk line marking devices;

(21) FIGS. 20-22 are sectioned sequential side views of the chalk wheel of the illustrative chalk line marking device, more particularly illustrating marking of the chalk lines on the bottom plate as the chalk wheel is rolled along the bottom plate;

(22) FIG. 23 is a top view, partially in section, of the foundation and the wall section on the foundation of the building with the chalk lines formed on the bottom plate at the selected chalk line spacings using the chalk wheel; and

(23) FIG. 24 is a side view, partially in section, of the wall section, with multiple wall studs extending upwardly from the bottom plate at the respective chalk lines and at wall stud spacings which correspond to the chalk line spacings of the chalk lines illustrated in FIG. 23.

(24) Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(25) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, and derivatives thereof shall relate to the subject matter as oriented in FIG. 1. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(26) All methods set forth in the present disclosure may be performed in any suitable order of steps

unless otherwise indicated herein or contradicted by the rules of logic. The use of any and all examples or exemplary language provided herein is intended to clearly describe the subject matter of the disclosure and is not intended to be limiting on the scope of the subject matter set forth in the claims. No element, step, ingredient, or limitation mentioned or described in the specification shall not be construed as regarding any unclaimed component, step, or limitation to be essential in practicing the claimed subject matter.

(27) Unless expressly or implicitly indicated otherwise, throughout the description and the appended claims, the terms “comprise”, “comprising”, “comprised of” and the like are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, and are equivalent to the phrase, “including but not limited to”. Each embodiment disclosed herein can comprise, consist essentially of, or consist of its particular stated element, step, ingredient, or limitation. As used herein, the transition terms “comprise”, “comprises”, “comprising”, “include”, “includes”, “including”, “is”, “has”, “having” or the like means “includes, but is not limited to, and allows for the inclusion of unspecified elements, steps, ingredients, or limitations, even in major amounts”. The transitional phrase “consisting of” excludes any element, step, ingredient, or limitation not specified. The transition phrase “consisting essentially of” shall limit the scope of the embodiment to the specified elements, steps, ingredients, or limitations and to those that do not materially affect the embodiment. Throughout the written description, drawings and claims appended hereto, unless otherwise noted, it shall be recognized and understood that each embodiment of the described, illustrated and claimed subject matter may comprise, consist essentially of, or consist of any component, element or combination of components or elements set forth herein.

(28) Unless otherwise noted using precise or limiting terminology, all numbers which express quantities of ingredients throughout the specification and claims are to be understood as being approximations of the numerical value cited to express the quantities of those ingredients. As used throughout the specification and claims, the term “about” has the meaning reasonably ascribed to it by a person skilled in the art when used in conjunction with a stated numerical value or range, i.e., denoting from the exact stated value or range to somewhat more or somewhat less than the stated value or range, from a deviation of from 0% with respect to the stated value or range to up to and including 20% of the stated value or range in either direction.

(29) Various illustrative embodiments of the disclosure are described herein. Variations on the described illustrative embodiments may become apparent to those of ordinary skill in the art in reading the specification, drawings and claims of the disclosure. Accordingly, the disclosure encompassed by the specification, claims and drawings includes all modifications and equivalents of the subject matter recited in the claims as permitted by applicable law. Additionally, any combination of the elements in all possible variations thereof is encompassed by the subject matter of the disclosure unless otherwise indicated herein.

(30) The term “or combinations thereof as used herein refers to all permutations and combinations of the listed items preceding the term. For example, “A, B, C, or combinations thereof is intended to include at least one of A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, CBA, BCA, ACB, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB, AAA, AB, BBC, AAABCCCC, CBBAAA, CABABB, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.

(31) Although the process-flow diagrams show a specific order of executing the process steps, the order of executing the steps may be changed relative to the order shown in certain embodiments. Also, two or more blocks shown in succession may be executed concurrently or with partial concurrence in some embodiments. Certain steps may also be omitted from the process-flow diagrams for the sake of brevity. In some embodiments, some or all the process steps shown in the process-flow diagrams can be combined into a single process.

(32) Referring initially to FIGS. 1-11 of the drawings, an illustrative embodiment of the chalk line marking devices, hereinafter device, of the disclosure is generally indicated by reference number 1. As illustrated in FIGS. 9-11 and will be hereinafter described, in typical application, the device 1 may be suitable for dispensing a powdered chalk material 58 to place chalk lines 59 at selected chalk line spacings 68 on a surface 62. For example and without limitation, in some applications, the surface 62 may be a bottom plate which is on a foundation 61 of a building 60. The chalk line spacings 68 may serve as a guide to proper positioning or placement of wall studs 70 along the bottom plate 62 at wall stud spacings 71 which correspond in length to the chalk line spacings 68 in the framing or construction of an interior or exterior wall in the building 60. It will be recognized and understood, however, that the device 1 may have numerous other applications in the construction and other industries.

(33) As illustrated in FIGS. 1-5, the device 1 may include a device housing 2. As illustrated in FIG. 5, the device housing 2 may have a housing interior 8. The housing interior 8 may be sized and configured to contain a supply of the powdered chalk material 58. As used herein throughout, "powdered chalk material", "chalk" and like terms include but are not limited to any type of powdered or particulate material which can be dispensed onto a surface to mark a location on the surface and may have any color and particle size depending on the desired application. A chalk marking assembly slot 16 may be provided in the device housing 2. A chalk marking assembly 24 may be selectively extendable from the housing interior 8 through the chalk marking assembly slot 16. As illustrated in FIGS. 1 and 2, the chalk marking assembly 24 may include an elongated, flexible main marking member 25. An elongated, flexible guide member 30 may be disposed in parallel, spaced-apart relationship to the main marking member 25. Multiple, spaced-apart, parallel center marking members 34 may span the main marking member 25 and the guide member 30 in perpendicular relationship to each. The center marking members 34 may be disposed at mark spacings 36 with respect to each other. The mark spacings 36 may correspond in length to the chalk line spacings 68 (FIG. 10), respectively, between the chalk lines 59 which will be made on the bottom plate 62 or other surface using the device 1. In some embodiments, the width spacing between the main marking member 25 and the guide member 30 of the chalk marking assembly 24 may be about 2¼".

(34) Each of the main marking member 25, the guide member 30 and the center marking members 30 of the chalk marking assembly 24 may include at least one segment of string, thread, twine, cord, or a combination thereof. As further illustrated in FIGS. 1 and 2, in some embodiments, the center marking members 34 of the chalk marking assembly 24 may include multiple pairs of spaced-apart, parallel, adjacent center marking members 34. The distance between the center marking members 34 of each center marking member 34 pair may correspond to the thickness of each wall stud 70 (FIG. 11) which will be erected on the bottom plate 62 in construction of the wall of the building 60, typically as will be hereinafter further described.

(35) In various embodiments, the mark spacings 36 between the adjacent center marking members 34 or between the adjacent pairs of the center marking members 34 of the chalk marking assembly 24 may have any of various lengths or spacings typically depending on the application of the device 1. For example and without limitation, in some embodiments, the mark spacings 36 may include 16-inch mark spacings. In other embodiments, the mark spacings 36 may include 24-inch mark spacings. In construction of the wall of the building 60 (FIG. 11), the 16-inch mark spacings 36 may facilitate placement of the wall studs 70 at 16-inch center wall stud spacings 71, whereas the 24-inch mark spacings 36 may facilitate placement of the wall studs 70 at 24-inch center wall stud spacings 71.

(36) As illustrated in FIGS. 1-4, in some embodiments, the device housing 2 of the device 1 may include a bottom housing portion 3; a top housing portion 4; a pair of spaced-apart side housing portions 5; a front housing portion 6; and a rear housing portion 7. As illustrated in FIG. 6, the chalk marking assembly slot 16 may extend through the front housing portion 6. At least one slot

seal **20** having a seal opening **21** may be disposed in the chalk marking assembly slot **16**. The slot seal **20** may be disposed in the chalk marking assembly slot **16**. The slot seal **20** may include a flexible, resilient, plastic or rubber grommet, O-ring or the like which retains the powdered chalk material **58** in the housing interior **8** of the device housing **2**. The slot seal **20** may sealingly engage the main marking member **25**, the guide member **30** and the center marking members **34** as the chalk marking assembly **24** traverses through the seal opening **21** in extended and retracted positions of the chalk marking assembly **24** with respect to the device housing **2**.

(37) At least one chalk opening **10** may extend through the top housing portion **4** and/or any other portion of the device housing **2**. A chalk opening door **11** may be selectively deployable in a closed door position (FIG. 7) to close the chalk opening **10** or in an open door position (FIG. 8) to open the chalk opening **10**, typically according to the knowledge of those skilled in the art. For example and without limitation, in some embodiments, the chalk opening door **11** may slidably engage the top housing portion **4**. A door tab **12** may protrude from the chalk opening door **11** for manual or digital manipulation of the chalk opening door **11** between the closed and open door positions.

(38) As further illustrated in FIGS. 1-4, in some embodiments, at least one engaging member **40** may terminate the distal marking member end **26** of the main marking member **25** and the distal guide member end **31** of the guide member **30** of the chalk marking assembly **24**, exterior to the device housing **2**. The engaging member **40** may facilitate engagement of the chalk marking assembly **24** with an anchor structure **65** (FIG. 9) to maintain the chalk marking assembly **24** in the extended position with respect to the device housing **2** in typical application of the device **1**, which will be hereinafter described. In some embodiments, the engaging member **40** may have a hook-shaped structure typically with a main member portion **41**, a proximal member portion **42** extending perpendicularly from the main member portion **41**, a distal member portion **43** extending perpendicularly from the proximal member portion **42** and a hook portion **44** extending perpendicularly from the distal member portion **43** toward the main member portion **41**. In other embodiments, the engaging member **40** may have alternative designs and configurations suitable for the purpose. In some embodiments, the engaging member **40** may include two hooks with $\frac{3}{4}$ offset.

(39) As illustrated in FIG. 5, a reel assembly **48** may be disposed in the housing interior **8** of the device housing **2**. As illustrated in FIGS. 1 and 2, a reel crank **54** may drivingly engage the reel assembly **48** for rotation. The chalk marking assembly **24** may be wound on the reel assembly **48** for selective deployment of the chalk marking assembly **24** between the retracted and extended positions with respect to the device housing **2** via rotation of the reel assembly **48** by actuation of the reel crank **54**. The chalk marking assembly **24** may have any design which is suitable for the purpose. Accordingly, as further illustrated in FIG. 5, in some embodiments, a reel axle **50** may be disposed in the housing interior **8** of the device housing **2**. The reel axle **50** may span the side housing portions **5** of the device housing **2**. A reel **49** may be journaled for rotation with or without respect to the reel axle **50**. The proximal ends (not illustrated) of the main marking member **25** and the guide member **30** may be attached to the reel **49** and wound thereon. Accordingly, by rotation of the reel **49** in a first direction, typically by manual manipulation of the reel crank **54**, the chalk marking assembly **24** may be wound in a first direction and retracted into the housing interior **8** through the chalk marking assembly slot **16**, as illustrated in FIG. 4. By rotation of the reel **49** in the opposite, second direction, the chalk marking assembly **24** may be wound in the second direction and extended from the housing interior **8** through the chalk marking assembly slot **16**, as illustrated in FIG. 3. In some embodiments, the reel **49** of the reel assembly **48** may be spring-loaded by a reel spring **51**. The reel spring **51** may be configured to normally bias the reel **49** in the winding direction and maintain the chalk marking assembly **24** in the retracted position (FIG. 4) in the housing interior **8**. The chalk marking assembly **24** may be grasped and pulled or extended from the housing interior **8** against the bias imparted by the reel spring **51**. A reel catch **52** may be configured to engage the reel **49** to maintain the chalk marking assembly **24** in the selected

extended position with respect to the device housing 2. The reel catch 52 may be released typically by pulling on the chalk marking assembly 24, by actuating a release lever (not illustrated) on the exterior of the device housing 2 and/or by initiating rotation of the reel crank 54 in the winding direction, typically according to the knowledge of those skilled in the art.

(40) As illustrated in FIGS. 9-11, in typical application, the device 1 may facilitate placement of the chalk lines 59 (FIG. 10) on the bottom plate 62. The chalk lines 59 may serve as a location guide for placement of the wall studs 70 on the bottom plate 62 in the construction of an interior or exterior wall in the building 60. Accordingly, the bottom plate 62 may extend along the surface of the foundation 61 from a vertical end stud 63 at one end of the wall.

(41) A supply of the powdered chalk material 58 may initially be placed into the housing interior 8 of the device housing 2 typically by deploying the chalk opening door 11 of the chalk opening 10 from the closed door position illustrated in FIG. 7 to the open door position illustrated in FIG. 8, after which the chalk opening door 11 may be closed. The chalk marking assembly 24 may next be deployed from the retracted position in the housing interior 8 of the device housing 2, as illustrated in FIG. 4, to a selected extended position from the device housing 2, as illustrated in FIG. 3. The length of the chalk marking assembly 24 which is extended from the device housing 2 may depend on the length of the marking area along the bottom plate 62 on which the chalk lines 59 will be placed. Accordingly, as illustrated in FIG. 9, in some applications, an anchor structure 65 may be erected, installed or placed on the bottom plate 62 at the end of the marking area on the bottom plate 62 which is opposite the end stud 63 and the engaging member 40 on the chalk marking assembly 24 engaged with the anchor structure 65. The device housing 2 may then be pulled horizontally away from the anchor structure 65 and toward the end stud 63 such that the chalk marking assembly 25 unwinds from the reel 49 (FIG. 5) of the reel assembly 48 and extends horizontally over and above the bottom plate 62 and from the housing interior 3 through the chalk marking assembly slot 16 typically until the device housing 2 is positioned adjacent to the end stud 63, as illustrated in FIG. 9. The anchor structure 65 may be any type of structure or object which is available on the construction site and with which the engaging member 40 can remain securely engaged as the chalk marking assembly 25 is extended from the device housing 2. For example and without limitation, in various applications, the anchor structure 65 may include a stud, box, toolbox or block of wood, for example and without limitation.

(42) In its retracted position in the housing interior 8, the chalk marking assembly 24 is submerged in the powdered chalk material 58. As the chalk marking assembly 24 is subsequently extended from the housing interior 8, a substantial quantity of the powdered chalk material 58 may remain coated on the main marking member 25, the guide member 30 and the center marking members 34 of the chalk marking assembly 24. Accordingly, as illustrated in FIG. 9, the extended chalk marking assembly 24 may be manually or digitally plucked or pulled upwardly and then released such that the main marking member 25 and the guide member 30 recoil downwardly and discharge powdered chalk material 58 from the main marking member 25, the guide member 30 and the center marking members 34 and onto the bottom plate 62. As illustrated in FIG. 10, the chalk lines 59 thusly formed on the bottom plate 62 may correspond to the pattern of the main marking member 25, the guide member 30 and each center marking member 34. The chalk lines 59 may therefore serve as a guide to subsequent placement of the wall studs 70 on the bottom plate 62, as illustrated in FIG. 11, with the lower end of each wall stud 70 typically placed within the rectangular confines of the chalk lines 59 formed by the main marking member 25, the guide member 30 and each pair of closely adjacent center marking members 34. The wall studs 70 may thus be placed and secured to the bottom plate 62 at the wall stud spacings 71 which correspond to the chalk line spacings 68. In some applications, the wall stud spacings 71 may be 16 center spacings whereas in other applications, the wall stud spacings 71 may be 24 inch or some other spacing.

(43) The same procedure may be carried out to facilitate placement of the wall studs 70 along the

entire length of the bottom plate **62** in framing of the wall. After each formation of the chalk lines **59**, and prior to subsequent use, the chalk marking assembly **24** may be deployed from the extended position back to the retracted position in the housing interior **8** of the device housing **2** to again coat the chalk marking assembly **24** with the powdered chalk material **58**. This may be accomplished by manual rotation of the reel crank **54** and/or by the biasing action of the reel spring **51** (FIG. 5).

(44) Referring next to FIGS. **12-24** of the drawings, an illustrative chalk wheel embodiment of the chalk line marking devices, hereinafter device, is generally indicated by reference number **100**. As illustrated in FIGS. **12-18**, the device **100** may include a chalk wheel handle **126**. A chalk wheel **101** may be supported by and journaled for rotation with respect to the chalk wheel handle **126**. The chalk wheel **101** may include a wheel housing **102**. The wheel housing **102** may include a circumferential outer wheel housing portion **104**. As illustrated in FIG. **14**, multiple chalk dispensing cavities **114** may be provided in the outer wheel housing portion **104**. Each chalk dispensing cavity **114** may be sized and configured to contain a supply of a powdered chalk material **158** and dispense the powdered chalk material **158** from the wheel housing **102** at the outer wheel housing portion **104** responsive to rolling of the chalk wheel **101** on the bottom plate **162** (FIG. **23**) or other surface to form the chalk lines **159** on the bottom plate **162**. As illustrated in FIG. **14**, the chalk dispensing cavities **114** may be disposed at mark spacings **134** with respect to each other along the outer wheel housing portion **104**. The mark spacings **134** may correspond in length to the chalk line spacings **168** or chalk line spacing **168** pairs (FIG. **23**) which may be formed on the bottom plate **162** in construction of the wall in the building **160** in application of the device **100**, as will be hereinafter described.

(45) As further illustrated in FIG. **14**, in some embodiments, the chalk dispensing cavities **114** may include multiple pairs of spaced-apart, adjacent chalk dispensing cavities **114**. The distance between the chalk dispensing cavities **114** of each chalk dispensing cavity **114** pair may correspond to the thickness of each wall stud **170** (FIG. **24**) which will be erected on the bottom plate **162** in construction of the wall of the building **160**, typically as will be hereinafter further described.

(46) In various embodiments, the mark spacings **134** between the adjacent chalk dispensing cavities **114** or between the adjacent pairs of the chalk dispensing cavities **114** may have any of various lengths or spacings typically depending on the application of the device **100**. For example and without limitation, in some embodiments, the mark spacings **134** may include 16-inch mark spacings, with the wheel housing **102** having a circumference of 32". In other embodiments, the mark spacings **134** may include 24-inch mark spacings, with the wheel housing **102** having a circumference of 48". In construction of the wall of the building **160** (FIG. **24**), the 16-inch mark spacings **134** may facilitate placement of the wall studs **170** at 16-inch center wall stud spacings **171**, whereas the 24-inch mark spacings **134** may facilitate placement of the wall studs **170** at 24-inch center wall stud spacings **171**.

(47) As illustrated in FIGS. **12-14**, the wheel housing **102** of the chalk wheel **101** may include a wheel axle **108**. A pair of spaced-apart side wheel housing portions **103** may be rotatably mounted with respect to the wheel axle **108**. The outer wheel housing portion **104** may extend between the side wheel housing portions **103**. In some embodiments, the width of the chalk wheel **101** between the side wheel housing portions **103** may be about 2¾". As illustrated in FIGS. **14** and **15**, a wheel housing interior **105** may be formed by and between the side wheel housing portions **103** and the outer wheel housing portion **104**. As illustrated in FIG. **12**, in some embodiments, a housing support frame **106** may be journaled for rotation with respect to the wheel axle **108** typically according to the knowledge of those skilled in the art. The side wheel housing portions **103** may be supported by the housing support frame **106**.

(48) As illustrated in FIGS. **12** and **15**, in some embodiments, at least one chalk opening **110** may be provided in the wheel housing **102** in communication with the wheel housing interior **105**. For example and without limitation, in some embodiments, the chalk opening **110** may be provided in

at least one of the side wheel housing portions **103** of the wheel housing **102**. At least one chalk opening door **111** may be selectively deployable in open and closed door positions with respect to the chalk opening **110**. The chalk opening **110** may facilitate placement of the powdered chalk material **158** into the wheel housing interior **105**.

(49) As illustrated in FIG. **16**, in some embodiments, an inner cavity slot **115** may establish communication between the wheel housing interior **105** of the wheel housing **102** and each chalk dispensing cavity **114**. An outer cavity slot **116** may communicate with each chalk dispensing cavity **114**. The outer cavity slot **116** may open to the outer surface of the outer housing portion **104** of the wheel housing **102**. Accordingly, the inner cavity slot **115** and the outer cavity slot **116** may facilitate flow of the powdered chalk material **158** from the wheel housing interior **105** through the chalk dispensing cavity **114** to the exterior of the outer wheel housing portion **104**. As illustrated in FIGS. **17** and **18**, in some embodiments, each chalk dispensing cavity **114** may have cavity ends **117** which may open to the respective side wheel housing portions **103** of the wheel housing **102** typically for purposes which will be hereinafter described.

(50) As further illustrated in FIGS. **16-18**, in some embodiments, a chalk dispensing insert **118** may be deployed in each chalk dispensing cavity **114**. Each chalk dispensing insert **118** may be sized and configured to contain the powdered chalk material **158** and dispense the powdered chalk material **158** from the wheel housing **102** at the outer wheel housing portion **104**. As illustrated in FIG. **16**, each chalk dispensing insert **118** may include a typically elongated, cylindrical insert wall **119** forming an insert interior **120**. An inner insert slot **121** and an outer insert slot **122** may extend through the insert wall **119** in opposite or diametrically opposed relationship to each other. As further illustrated in FIG. **16**, the inner insert slot **121** may align or register with the inner cavity slot **115**, and the outer insert slot **122** may align or register with the outer cavity slot **116**, of the chalk dispenser cavity **114**. Accordingly, in typical application of the device **100**, which will be hereinafter described, the powdered chalk material **158** may flow from the wheel housing interior **105** through the inner cavity slot **115**, the inner insert slot **121**, the insert interior **120**, the outer insert slot **122** and the outer cavity slot **116**, respectively, to the exterior of the outer wheel housing portion **104** to form the chalk lines **159** (FIG. **23**) on the bottom plate **162** in construction of the wall of a building **160**.

(51) In some embodiments, the inner cavity slot **115** may be omitted from the outer wheel housing portion **104** of the wheel housing **102**. The inner insert slot **121** may be omitted from the insert wall **119** of the chalk dispensing insert **118**. The insert interior **120** of each chalk dispensing insert **118** may be pre-filled with the powdered chalk material **158**, which may be dispensed from the insert interior **120** through the outer insert slot **122** and the outer cavity slot **116**, respectively, at the surface of the outer wheel housing portion **104**.

(52) As illustrated in FIG. **18**, in some embodiments, each chalk dispensing insert **118** may be releasably deployed in its corresponding chalk dispensing cavity **114**. Accordingly, the chalk dispensing insert **118** may be inserted into and removed from the chalk dispensing cavity **114** through one of the open cavity ends **117** of the chalk dispensing cavity **114**. Chalk dispensing inserts **118** which contain powdered chalk material **158** having different colors, sizes, textures and the like may be interchangeably deployed in the chalk dispensing cavities **114** to form chalk lines **159** having various colors for different marking purposes depending on the application.

(53) The chalk wheel handle **126** of the device **100** may have any design which is suitable for the purpose of guiding the chalk wheel **101** as the chalk wheel **101** is rolled in a substantially straight line along the bottom plate **162**. As illustrated in FIGS. **12** and **13**, in some embodiments, the chalk wheel handle **126** may include a proximal handle segment **127**. The proximal handle segment **127** may extend from the wheel axle **108** of the chalk wheel **101**. A distal handle segment **128** may extend from the proximal handle segment **127**. A handle grip **130** may extend from the distal handle segment **128**. A handle hinge **129** may pivotally attach the distal handle segment **128** to the proximal handle segment **127**. Accordingly, the chalk wheel handle **126** may be selectively

deployable in an extended position, as shown, to facilitate application of the device **100**. The chalk wheel handle **126** may be selectively deployable in a folded position (not illustrated) to facilitate space-efficient storage or transport of the device **100** by pivoting the distal handle segment **128** with respect to the proximal handle segment **127** at the handle hinge **129**.

(54) As illustrated in FIGS. **19-24**, in typical application, the device **1** may facilitate placement of the chalk lines **159** on the bottom plate **162** to serve as a location guide for placement of the wall studs **170** (FIG. **24**) on the bottom plate **162** in the construction of the wall in the building **160**. A supply of the powdered chalk material **158** may initially be placed into the wheel housing interior **105** (FIG. **14**) of the wheel housing **102** of the chalk wheel **101** typically by deploying the chalk opening door **111** of the chalk opening **110** from the closed door position to the open door position, pouring the powdered chalk material **158** through the chalk opening **110** and then closing the chalk opening door **111**. Alternatively, the pre-filled chalk dispensing inserts **118** may be placed in the respective chalk dispensing cavities **114**, typically as was heretofore described.

(55) As illustrated in FIG. **19**, the chalk wheel **102** may be placed on the bottom plate **162** at the beginning of the marking area along the bottom plate **162** on which the chalk lines **159** will be placed, typically either away from and facing toward the end stud **163**, as illustrated, or adjacent to and facing away from the end stud **163**. The operator (not illustrated) of the device **100** may stand on the bottom plate **162** or on the foundation **161** next to the bottom plate **162** and grasp the handle grip **130** of the chalk wheel handle **126**. As illustrated in FIGS. **20-22**, the chalk wheel **101** may be rolled along the bottom plate **162** toward or away from the end stud **163** along the marking area such that the powdered chalk material **158** is discharged typically from the chalk dispensing inserts **118** through the respective outer insert slots **122** and registering outer cavity slots **116** (FIG. **16**) onto the bottom plate **162** as the outer cavity slots **116** contact the surface of the bottom plate **162**, as illustrated in FIG. **22**, forming the chalk lines **159** on the bottom plate **162**. The chalk lines **159** may therefore serve as a guide to subsequent placement of the wall studs **170** on the bottom plate **162**, as illustrated in FIG. **24**, with the lower end of each wall stud **170** typically placed between the chalk lines **159** formed by the paired adjacent chalk dispensing inserts **118**. The wall studs **170** may thus be placed and secured to the bottom plate **162** at the wall stud spacings **171** which correspond to the chalk line spacings **168**. In some applications, the mark spacings **134**, and thus, the wall stud spacings **171**, may be 16 center spacings whereas in other applications, the mark spacings **134** and the wall stud spacings **171** may be 24 inch or some other spacing.

(56) The same procedure may be carried out to facilitate placement of the wall studs **170** along the entire length of the bottom plate **162** in framing of the wall. After each formation of the chalk lines **159**, and prior to subsequent use, a replacement supply of the powdered chalk material **158** may be placed in the wheel housing interior **105** of the wheel housing **102** typically through the chalk opening **110**. Alternatively, one or more of the typically pre-filled chalk dispensing inserts **118** may be replaced in one or more of the respective chalk dispensing cavities **114**. The chalk wheel **101** may be rolled along the bottom plate **162** to again form the chalk lines **159** along a marking area on another segment of the bottom plate **162**.

(57) While certain illustrative embodiments of the disclosure have been described above, it will be recognized and understood that various modifications can be made to the embodiments and the appended claims are intended to cover all such modifications which may fall within the spirit and scope of the disclosure.

Claims

1. A chalk line marking device suitable for placing chalk lines at selected chalk line spacings on a surface, the device comprising: a device housing; a housing interior in the device housing, the housing interior sized and configured to contain a supply of a powdered chalk material; a chalk marking assembly slot in the device housing; a chalk marking assembly selectively extendable

from the housing interior through the chalk marking assembly slot, the chalk marking assembly comprising: an elongated, flexible main marking member; an elongated, flexible guide member disposed in parallel, spaced-apart relationship to the main marking member; and a plurality of center marking members spanning the main marking member and the guide member, the plurality of center marking members disposed at mark spacings with respect to each other, the mark spacings corresponding in length to the chalk line spacings, respectively.

2. The chalk line marking device of claim 1 further comprising a reel assembly disposed in the housing interior of the device housing and a reel crank engaging the reel assembly for rotation, and wherein the chalk marking assembly is wound on the reel assembly.

3. The chalk line marking device of claim 1 wherein each of the main marking member and the guide member of the chalk marking assembly is a segment of string, thread, twine, cord or a combination thereof.

4. The chalk line marking device of claim 1 wherein the mark spacings comprise 16 inch mark spacings.

5. The chalk line marking device of claim 1 wherein the mark spacings comprise 24 inch mark spacings.

6. The chalk line marking device of claim 1 further comprising at least one chalk opening in the device housing and at least one chalk opening door selectively deployable in open and closed door positions with respect to the at least one chalk opening.

7. The chalk line marking device of claim 1 further comprising at least one slot seal in the chalk marking assembly slot, and wherein the at least one slot seal sealingly engages the chalk marking assembly.

8. The chalk line marking device of claim 1 further comprising at least one engaging member on the main marking member and the guide member of the chalk marking assembly.

9. The chalk line marking device of claim 1 wherein the plurality of center marking members comprises a plurality of center marking member pairs, respectively.

10. A chalk line marking device suitable for placing chalk lines at selected chalk line spacings on a surface, the device comprising: a chalk wheel handle; and a chalk wheel carried by and journaled for rotation with respect to the chalk wheel handle, the chalk wheel comprising: a wheel housing comprising a circumferential outer wheel housing portion; and a plurality of chalk dispensing cavities in the outer wheel housing portion, each of the plurality of chalk dispensing cavities sized and configured to contain a supply of a powdered chalk material and dispense the powdered chalk material from the wheel housing at the outer wheel housing portion responsive to rolling of the chalk wheel on the surface, the plurality of chalk dispensing cavities disposed at mark spacings with respect to each other along the outer wheel housing portion, the mark spacings corresponding in length to the chalk line spacings, respectively.

11. The chalk line marking device of claim 10 further comprising a plurality of chalk dispensing inserts in the plurality of chalk dispensing cavities, respectively, the plurality of chalk dispensing inserts sized and configured to contain the powdered chalk material and dispense the powdered chalk material from the wheel housing at the outer wheel housing portion.

12. The chalk line marking device of claim 11 wherein the plurality of chalk dispensing inserts is releasably disposed in the plurality of chalk dispensing cavities, respectively.

13. The chalk line marking device of claim 10 further comprising a wheel housing interior in the wheel housing, the wheel housing interior sized and configured to contain the powdered chalk material and dispense the powdered chalk material to the plurality of chalk dispensing cavities, respectively.

14. The chalk line marking device of claim 13 further comprising at least one chalk opening in the wheel housing and communicating with the wheel housing interior and at least one chalk opening door selectively deployable in open and closed door positions with respect to the at least one chalk opening.

15. The chalk line marking device of claim 10 wherein the mark spacings comprise 16 inch mark spacings.

16. The chalk line marking device of claim 10 wherein the mark spacings comprise 24 inch mark spacings.

17. The chalk line marking device of claim 10 wherein the plurality of chalk dispensing cavities comprises a plurality of chalk dispensing cavity pairs, respectively.

18. A chalk line marking device suitable for placing chalk lines at selected chalk line spacings on a surface, the device comprising: a chalk wheel handle; and a chalk wheel carried by and journaled for rotation with respect to the chalk wheel handle, the chalk wheel comprising: a wheel housing comprising a pair of spaced-apart side wheel housing portions, a circumferential outer wheel housing portion on the side wheel housing portions and a wheel housing interior formed by the side wheel housing portions and the outer wheel housing portion; a plurality of chalk dispensing cavities in the outer wheel housing portion, the plurality of chalk dispensing cavities disposed at mark spacings with respect to each other, the mark spacings corresponding in length to the chalk line spacings, respectively; a plurality of inner cavity slots establishing communication between the wheel housing interior and the plurality of chalk dispensing cavities, respectively; a plurality of outer cavity slots communicating with the plurality of chalk dispensing cavities, respectively, and opening to the outer wheel housing portion; and a plurality of chalk dispensing inserts in the plurality of chalk dispensing cavities, respectively, and communicating with the plurality of inner cavity slots and the plurality of outer cavity slots, respectively, the plurality of chalk dispensing inserts sized and configured to contain a supply of powdered chalk and dispense the powdered chalk from the wheel housing through the plurality of outer cavity slots, respectively, at the outer wheel housing portion.

19. The chalk line marking device of claim 18 wherein the mark spacings comprise 16 inch mark spacings.

20. The chalk line marking device of claim 18 wherein the mark spacings comprise 24 inch mark spacings.
